

IT Support Troubleshooting & Incident Resolution Lab
Practical IT Support Home Lab

Prepared by: Alanoud Alfouzan

Incident #001 — Network Connectivity Troubleshooting

Issue: Network connectivity verification

Tools/Commands: ip addr, ping, ip route

Result: Network interface, gateway connectivity, internet access, and DNS resolution were successfully verified.

Screenshots:

Screenshot 1 – Network Interface Configuration

Screenshot 2 – Gateway Connectivity Test

Screenshot 3 – Internet Connectivity Test

Screenshot 4 – DNS Resolution and Connectivity Test

Screenshot 5 – Routing Table Configuration

Incident #002 — DNS Resolution Troubleshooting

Issue: DNS resolution verification

Objective: Verify the DNS configuration and confirm that domain names can be resolved successfully.

Tool/Command: cat /etc/resolv.conf

Finding: The system is configured to use DNS server 192.168.8.1.

Result: DNS resolution was successfully verified using ping google.com.

Screenshots:

Screenshot 6 – DNS Configuration – /etc/resolv.conf

Incident #003 – Slow Computer Performance

Issue: Slow computer performance

Objective: Check system memory usage, disk space, and CPU activity to identify potential performance issues.

Tools/Commands:

free -h

df -h

top

Findings:

The system has 1.3 GiB of available memory, with no swap memory currently in use.

The main disk has 54 GB of available space, with 28% of the storage currently in use.

CPU idle time was 99.0%, with a low load average of 0.19, 0.17, and 0.13.

No zombie processes were detected.

Result:

Memory, disk space, and CPU activity were within normal ranges during the check. No clear resource-related cause for slow performance was identified.

Screenshots:

Screenshot 7 – Memory Usage Check

Screenshot 8 – Disk Space Check

Screenshot 9 – CPU and Process Activity Check

Incident #004 – IP Configuration Verification

Issue:

IP configuration verification

Objective:

Verify the IP address, network interface, routing configuration, and gateway connectivity.

Tools/Commands:

ip addr show eth0

ip route

ping -c 4 10.0.2.2

Findings:

Network interface eth0 is UP. •

IP address: 10.0.2.15/24 •

Broadcast address: 10.0.2.255 •

Default gateway: 10.0.2.2 •

Network: 10.0.2.0/24 •

Gateway connectivity: 4 packets received with 0% packet loss. •

Average response time: 0.252 ms. •

Result:

IP configuration, routing, and gateway connectivity were successfully verified. No configuration issue was identified.

Screenshots:

Screenshot 10 – IP Configuration Check

Screenshot 11 – Gateway Connectivity Verification

Incident #005 – Application Process Verification

Issue:

Application process verification and basic process management.

Objective:

Verify running application processes, monitor system activity, and practice basic process management.

Tools/Commands:

```
ps aux --sort=-%cpu | head
ps aux | grep -i firefox
firefox
ps -p 4573 -o pid,stat,etime,cmd
kill 4573
```

Findings:

System processes were reviewed and CPU usage was low.

Firefox was successfully launched and verified.

The Firefox process was not running after the application was closed.

A background process was created and successfully identified using PID 4573.

The process was successfully terminated using the kill command.

Result:

Application processes were successfully monitored and managed. The test confirmed the ability to identify and terminate a running process.

Screenshots:

Screenshot 12 – Running Processes and CPU Usage

Screenshot 13 – Application Process Verification

Screenshot 14 – Background Process Verification

Screenshot 15 – Process Termination

Incident #006 – Disk Space Issue

Issue:

Disk space usage verification.

Objective:

Check available disk space and identify directories or log files that may be consuming significant storage.

Tools/Commands:

`df -h`

`du -sh ~/*`

`du -sh /var/log/*`

Findings:

Main disk `/dev/sda1` has 78 GB total space.

21 GB is used and 54 GB is available.

Disk usage is 28%.

The largest directory in the user profile is Downloads at 74 MB.

`/var/log/journal` uses 465 MB.

Some log directories returned Permission denied because they require elevated privileges.

Result:

Disk space was successfully assessed. The system has sufficient available storage, and no immediate disk space issue was identified.

Screenshots:

Screenshot 16 – Disk Space Overview

Screenshot 17 – Disk Space Usage by Directory

Screenshot 18 – System Log Storage Analysis

Incident #007 – User Permissions Issue

Issue:

File permissions verification and configuration.

Objective:

Verify file permissions and apply appropriate access restrictions to a user file.

Tools/Commands:

```
ls -l ~/test_permissions.txt  
chmod 600 ~/test_permissions.txt  
stat -c "%A %U:%G %n" ~/test_permissions.txt
```

Findings:

The file was initially set to -rw-r--r--.

The file owner was kali.

Permissions were changed using chmod 600.

The final permissions were -rw-----.

The final verification confirmed that only the owner has read and write permissions.

Result:

File permissions were successfully reviewed and restricted to the file owner, demonstrating basic Linux access control and permission management.

Screenshots:

Screenshot 19 – File Permissions Check

Screenshot 20 – Updated File Permissions

Screenshot 21 – Permission Verification

Incident #008 – Basic Security Incident

Issue:

Basic security verification of network services and SSH configuration.

Objective:

Identify listening network services, verify the associated processes, review the effective SSH configuration, and confirm that the SSH service is running correctly.

Tools/Commands:

```
ss -tuln
```

```
sudo ss -tulpn
```

```
sudo sshd -T | grep -E "^(permitrootlogin|passwordauthentication|port) "
```

```
sudo systemctl status ssh --no-pager
```

Findings:

SSH is listening on port 22 through IPv4 and IPv6.

PostgreSQL is listening on port 5432 and is restricted to localhost.

Port 22 is associated with the sshd process.

Port 5432 is associated with the postgres process.

Effective SSH configuration shows:

Port: 22

Root login: without-password

Password authentication: enabled

SSH service status is **active (running)**.

SSH configuration validation completed successfully with status 0/SUCCESS.

Result:

Network services and SSH configuration were successfully reviewed. Listening ports and associated processes were identified, SSH access settings were verified, and the SSH service was confirmed to be running correctly. No immediate security issue was identified during the basic verification.

Screenshots:

Screenshot 22 – Listening Ports and Network Services

Screenshot 23 – Listening Services Identification

Screenshot 24 – Effective SSH Configuration

Screenshot 25 – SSH Service Status

Project Summary

Project Title:

IT Support Troubleshooting & Incident Resolution Lab

Project Type:

Practical IT Support Home Lab

Project Objective:

To practice troubleshooting and resolving common IT support incidents, including network connectivity, DNS, system performance, application processes, disk usage, file permissions, and basic security configuration.

Project Scope:

The project covered eight practical IT support incidents using a Kali Linux virtual environment. Each incident was investigated using command-line tools, system checks, configuration verification, and basic troubleshooting techniques.

Key Areas Covered:

Network connectivity and IP configuration

DNS resolution

System performance monitoring

Application process management

Disk space analysis

File permissions and access control

Network services and listening ports

SSH configuration and service verification

Outcome:

The project demonstrated practical skills in troubleshooting, system analysis, configuration verification, process management, and incident documentation.

Tools & Skills Used

Tools:

Kali Linux

Oracle VM VirtualBox

Linux Terminal

Network and system utilities: ip, ping, ss, ps, top, free, df, du, chmod, stat
systemctl

Skills Demonstrated:

Network troubleshooting

IP and DNS verification

System performance monitoring

Process identification and management

Disk space analysis

Linux file permissions

Network service and SSH verification

Technical incident documentation

Project Outcome

This project provided practical experience in investigating and troubleshooting common IT support incidents using a structured approach.

The project strengthened practical skills in network troubleshooting, Linux system checks, process management, file permissions, basic security verification, and technical incident documentation.

Each incident was documented with the commands used, findings, results, and supporting screenshots.